Ulaganathan K B

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.47
Thamarai International School	Class XII (CBSE)	2023	92.40%
Thamarai International School	Class X (CBSE)	2021	97.00%

INTERNSHIP

Samsung Research & Development Institute - Delhi | 2-Month Internship

- Enabled **cross-platform deployment** of an **AI Agent**.
- Created a robust **Agent Wrapper** enabling integration with **Non-OpenAI-compatible LLM**.
- **Linked** the agent with **internal tools** while complying with company restrictions.
- Reduced the companion app's footprint from **>500 MB to ~75 MB** while improving robustness.

PROJECTS

Project: ECG to Time Series Converter Prof: Nagarajan

- Implemented an end-to-end AI pipeline for extracting **1D time-series data** from **realistic ECG images**.
- Developed a **PowerShell script** using **ECG generator kit** to build training batches in realistic and B&W formats.
- Used **Object Detection model** (Detectron2 / Faster R-CNN) to detect and **crops standard ECG leads**.
- Used **DL Segmentation model** to **isolate ECG signals** from gridlines and background noise.
- Developed a PyTorch-based custom model using **Cosine Annealing & Huber Loss** to extract 1D time-series data.

Project: Neural Operators for Linearised Magnetohydrodynamic Wave Simulation

Course: Machine Learning for Physical Sciences **Prof:** Sangkha Borah

- Used **FNO & PINO** to create **resolution-invariant surrogate models** to approximate 2D L-MHD solution operator**.**
- Achieved **<2% relative L2-Error** on dominant Alfvén wave modes(uz,Bz) and **<8% Mean Relative Error** overall.
- Inference time of **~5.5×105 s/sample**--orders of magnitude faster than CPU-bound PLUTO finite-volume runs.
- Used **PLUTO Solver** with C/Python to generate **100k 32×32 input output pairs** across randomized parameters.
- Developed a **NumPy preprocessing** pipeline with peak RAM usage under **40 MB** for an **800 MB dataset**.
- Proved that **incorporating physics constraints reduced violations by >50%,** suppressed error compounding over autoregressive rollouts and enabled graceful accuracy degradation when extrapolating beyond training time.

Project: Arbitrary-Precision Arithmetic Library

Course: Software Development Fundamentals **Prof:** Ashish Mishra

- Developed a Java library for **arbitrary-precision integer and floating-point arithmetic** supporting addition, subtraction, multiplication, and division using OOP principles.
- The library handles the input by storing it as an array of digits ensuring precision without overflow.
- **No rounding off** is done (Except for FloatingDiv which is rounded off after 30 decimal places **similar to BigDecimal**).
- Used Python script to automate execution via CLI.

SCHOLASTIC ACHIEVEMENTS

- Cleared **NTSE Level 2**.
- Took part in **VVM State Level Camp (Level II) Examination,** Tamil Nadu 2019-20.
- Qualified for **All India SPOT PRELIMS** 2019-20 after clearing **SPOT SENIOR** held by VSSF.

- Participated in various **ASSET Exams** and **Olympiads** like IMO (five consecutive years), IEO, NCO, NSO, Green Olympiad and Tamil Olympiad multiple times throughout my school life.
- Chosen to represent my school in **Zonal Level Spelling Bee** and took part in school level Spelling Bee multiple times.

SKILLS

- **Programming:** Python, C++
- **Tools:** Kaggle, Colab, VS Code, PyCharm, Jupyter Notebook, Overleaf, Solid Edge, GitHub, Vim, Bash, PostgreSQL
- **Familiar With:** HTML, CSS, C, Java, OCaml, Bash/Shell, Linux, SQL

RELEVANT COURSES

CS: Data Mining, Database Management Systems, Linear Optimisation, Software Development Fundamentals, Theory of Computation, Algorithms, Discrete Maths, Operating Systems, Data Structure & Applications, Data Science Analysis

AI: AI in Finance, Foundations of Machine Learning, Machine Learning for Physical Sciences

Mathematics: Group Theory, Complex Variables, Statistics, Transform Techniques, Linear Algebra, Differential Equations, Calculus, Probability, Numerical Methods, Mathematical Physics, Applied Mathematics in Chemical Engineering.

Economics: Macroeconomics, Principles of Economics, Financial Institutions & Markets, Public Finance & Public Policy

EXTRACURRICULAR

- Completed all **8 Levels in Abacus** due to my Interest in Mathematics.
- Consistent **Gym** goer for the past 3 years.
- Like playing **Chess** and participated in school and district level Chess tournaments.
- In a bid to improve my Linguistic knowledge, completed till **Rastrabhasha** in Hindi Prachar Sabha.
- **Brown belt** holder in Karate.
- Won various awards in school for Academics and Sports